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Stock Index Price Prediction Using Sentimental Analysis

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I declare that this dissertation is all my own work, except as indicated in the text

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Abstract

This study tackles the difficulty of forecasting the Dow Jones Industrial Average (DJIA) index price by combining sentiment analysis with conventional financial prediction techniques. Anticipating movements in a stock index is inherently more complicated than predicting individual stock prices, as it involves compiling data from numerous companies, each exerting different degrees of influence. This complexity demands advanced models that can effectively capture and interpret overall market sentiment.

To tackle this challenge, we incorporate investor sentiment obtained from financial news headlines into a sophisticated Long Short-Term Memory (LSTM) neural network model. The research examines DJIA price fluctuations from 2022 to 2023, applying feature engineering techniques like volume-weighted sentiment, exponentially weighted moving averages (EWMA), and sentiment volatility to discern both short-term and long-term trends in market sentiment. We gather sentiment data from multiple reputable financial news sources to ensure a well-rounded and impartial perspective.

The sentiment-enhanced LSTM model outperformed the baseline, demonstrating improved prediction accuracy through lower Root Mean Square Error (RMSE) and higher R-squared (R^2) values. Key findings indicate that aggregated market sentiment and engineered temporal features significantly contribute to predicting stock index movements more effectively.

However, limitations include potential oversimplification of investor sentiment through news headlines and challenges in effectively weighting sentiment from individual companies. Future research should consider incorporating social media sentiment, exploring more advanced model architectures such as transformers, and refining feature weighting techniques.

This study contributes to financial forecasting literature by demonstrating the value of integrating qualitative sentiment data with quantitative indicators, offering a more robust approach to stock index prediction.

Keywords: Sentiment Analysis, Stock Index Prediction, Dow Jones Industrial Average, Machine Learning, LSTM, Financial Forecasting

Introduction

Background and Context

Stock price forecasting is a very difficult task that touches the interests of a wide range of stakeholders. Precise models of stock index prediction bear prime importance, as they enable investors to make better-informed decisions regarding their investment strategies and facilitate proper risk management. These models enable institutional investors and hedge funds to enhance portfolio performance and achieve a competitive advantage. Individual investors may benefit from the fact that their investment decisions will be made in a way that can better reflect what is expected from the market, therefore improving their financial outcomes. Several financial analysts and market consultants use predictive models for market signal analysis, forecasting future trends, and advising clients on strategic issues. This enables them to invest in opportunities and advise on the distribution of assets. Policymakers and regulatory authorities use stock index forecasts to assess the general economic condition for the detection of potential financial bubbles or downtrends and for the timely adoption of measures necessary for maintaining economic stability.

Accurate forecasts are a precept for financial stability, guiding monetary and fiscal policy actions that are geared at reducing economic risks. This calls for strong predictive models inspired by the mere volatility and complexity innate in financial markets. Financial markets are driven by many interacting factors, which inherently make them unpredictable and change quickly. Scarcely accurate predictions lessen uncertainty, improve efficiency in markets, and add to stability within the financial systems. This is especially the case when one considers an increasingly interdependent world economy that is best served by minimal systemic risk and encourages sustainable economic growth. Therefore, effective development of stock index prediction models is not only desirable but also vital for the benefit of the financial ecosystem.

The Dow Jones Industrial Average (DJIA) consists of 30 of the largest companies in nearly every sector of the US economy and is a leading indicator of the health of the nation's economy. It is also extremely hard to forecast, for it is a variable mix of what has been happening at different companies in the greater economic scene and globally. Since its inception in 1896, the DJIA has distinctly seen ups and downs influenced by certain historical events. For instance, the stock market crash in 1929 led to the Great Depression, when the DJIA fell almost 90% from its peak and barely recovered that peak in over two decades. On

another occasion, Black Monday in 1987, the DJIA plummeted 22.6%, indicating a fragile market prone to a shock. More recently, events such as the 2008 financial crisis—triggered by the collapse of the housing market and major financial institutions, as cited by Brunnermeier (2009)—and the COVID-19 pandemic in 2020, which caused unprecedented market volatility, have significantly adjusted the DJIA. These events drew special attention to the importance of trying to understand quantitative data and qualitative factors like investor sentiment in advance.

The performance of the DJIA is pegged onto a number of key economic indicators, such as GDP growth, unemployment rates, inflation, and interest rates. When the nation is experiencing solid growth in GDP and low unemployment figures, this often sends the DJIA upward on the back of investor confidence. Very high inflation levels or even an economic recession may cause market downturns. For example, the high-risk lending practices combined with the bursting of the housing bubble led to the 2008 financial crisis, resulting in a severe recession and precipitous decline in practically all stock market indices worldwide (Reinhart & Rogoff, 2009). In addition, geopolitical events and policy decisions can lead to sudden changes in investor mood, further complicating any prediction efforts. Sudden political reversals, international conflicts, or shifts in trade policies fuel market volatility whereby investors continuously readjust to new information.

Adding to these macroeconomic influences, the sentiment regarding individual companies within the DJIA becomes crucial to the index's performance. Other than traditional macroeconomic analysis, which considers quantitative indicators such as GDP and unemployment, sentiment analysis looks into qualitative data such as investor perceptions and reactions, sometimes from news releases and social media to earnings reports. The approach enables the observation of a psychological and emotional issue that takes over the market. Market perceptions of investors about a company's future are based on news releases, earnings reports, leadership changes, or product developments. Thus, huge fluctuations in the price may be observed for that particular company's stock. Because the DJIA is a price-weighted index, it would be very vulnerable to changes in the stock prices of constituent firms, especially those with higher stock price levels. For example, positive news about one of the leading technology companies in the DJIA may lead to rising investor confidence and upward movements in the index itself. On the other hand, negative news in the form of corporate scandals or regulatory causes lost trust, and declines set in. Hence, the DJIA moves with aggregate sentiment towards the individual company. This reflects how micro

sentiments may impact the macro dynamics of the market. Large classic quantitative models, such as various time series forecasting techniques and econometric models, usually cannot capture more qualitative factors that drive market fluctuations, like investor sentiment and surprise exogenous shocks. The inability to rely only on historical numerical data has engendered interest in incorporating sentiment analysis of financial news to enhance the predictive accuracy of such models.

Investor sentiment also tends to be signaled through financial news reporting, social media, and other news information that can often instantly move stock prices. News headlines tend to convince the general public away from or toward certain investments, and consequently, market activities immediately respond to them. For instance, during the 2008 financial crisis, negative news about the collapse of banks and other financial institutions resulted in overall panic with unprecedented sell-offs in the stock market. The sentiment analysis across various news sources could allow the seizing of an almost complete overview of the market mood and yield more accurate predictions. In general, practical sentiment analysis is performed by quantifying the emotional tone within the textual content, classifying that content as positive, negative, or neutral, and then integrating such foresight with quantitative data.

Improvements in NLP and machine learning increasingly ease extracting and quantifying the sentiment from textual data. Sentiment scoring, topic modeling, and semantic analysis allow researchers to efficiently process voluminous amounts of unstructured text data. These, in turn, moved toward unsophisticated predictive models. A combination of such traditional quantitative analysis with qualitative sentiment information can allow market dynamics to be captured more effectively.

Integrating sentiments from multiple news sources carries many opportunities, but it is equally challenging: it captures diverse opinions and reduces bias associated with dependence on one source, introduces complexities related to data aggregation, weighting different sources, and handling conflicting sentiments. Different styles of reporting, time of publication, and editorial biases make news outlets inconsistent in the sentiment data. For example, during the 2008 crisis, one could easily find sensationalist headlines from some sources that fed investor fear, while other sources made more tempered analyses. Depending on how one approaches these challenges, various robust methods will be required to aggregate and standardize sentiment information to ensure that the latter is reliable and valuable within predictive models.

However, with financial markets being impacted by many factors and the sentiment varying according to the context and the events reported, it remains to be seen if sentiment analysis works in stock index prediction. The rapid swing in sentiment embraced through news or global impacts of events such as the 2008 financial crisis and the COVID-19 pandemic has resulted in unexpected agitations within the market. This is complex and stresses the tester's interest in and rigorous evaluation of sentiment-based predictive models for real-world applications. Therefore, while sentiment analysis does open new avenues of improvement for stock index predictions, the models need further refinement to understand their full capabilities and limitations.

Objective and Research Questions

Objective

This dissertation aims to explore the potential of integrating sentiment analysis of financial news headlines collected from diverse sources to improve stock market prediction accuracy. The research specifically focuses on enhancing forecasts for the Dow Jones Industrial Average (DJIA) by merging qualitative sentiment insights with traditional quantitative indicators. By adopting this hybrid approach, the study aspires to deepen the understanding of market dynamics and better capture complex financial trends. The central objective is to innovate upon existing predictive frameworks by effectively combining numerical data with sentiment-based qualitative elements, ensuring a more adaptive and holistic methodology for tackling stock market fluctuations.

1. **Refine Stock Index Forecasting through Sentiment Integration:** The research seeks to elevate the precision of Dow Jones Industrial Average (DJIA) stock market predictions by embedding sentiment analysis from financial news headlines into traditional forecasting models. This approach emphasizes creating a dynamic predictive tool that synergizes numerical financial indicators with qualitative data insights.
2. **Develop a Comprehensive Sentiment Analysis Framework for Financial News:** Establish a robust framework for collecting, processing, and quantifying sentiment from financial news headlines across multiple reputable sources. This includes

ensuring data quality, reliability, and consistency to accurately reflect market sentiment over time.

3. **Examine the Effect of Multi News Source Sentiment Aggregation on Prediction Reliability:** Investigate how aggregating sentiment data from multiple financial news sources impacts the robustness and reliability of stock index predictions. This objective aims to understand whether combining sentiments from various news sources minimizes biases and enhances the model's generalizability.
4. **Design Advanced Deep Learning Models Integrating Sentiment and Market Data:** Create and optimize advanced deep learning models, particularly Long Short-Term Memory (LSTM) networks, that effectively integrate sentiment features with traditional market indicators. The goal is to capture complex temporal dependencies and improve the model's ability to predict stock index movements.
5. **Evaluate the Contribution of Sentiment Analysis to Model Performance:** Assess the extent to which incorporating sentiment analysis enhances the performance of stock index prediction models compared to using quantitative data alone. This involves comparing models with and without sentiment features using metrics like RMSE, MAE, and R-squared to quantify improvements.
6. **Investigate Methods for Weighting Individual Company Sentiment within the Index:** Explore approaches for appropriately weighing the sentiment of individual companies in the DJIA, considering factors like market capitalization or trading volume. This objective seeks to ensure that the aggregated sentiment accurately reflects each company's influence on the overall index value.

Research Questions

RQ1: How can financial news headlines be effectively scraped and processed to create a reliable and repeatable sentiment analysis pipeline for stock market prediction?

This question seeks to develop an efficient workflow for scraping and processing news data, ensuring a reliable pipeline that minimizes errors and inconsistencies in sentiment data collection.

RQ2: What sentiment-related features best capture the temporal relationship between financial news sentiment and DJIA stock index price movements?

This question aims to identify key sentiment features that effectively represent both short-term and long-term market reactions, contributing to better prediction outcomes.

RQ3: Does aggregating sentiment from multiple financial news sources lead to improved predictive accuracy compared to using single-source sentiment data?

This question investigates whether combining sentiment data from multiple news sources enhances prediction robustness and reduces biases inherent in single-source analyses.

RQ4: How can deep learning models, particularly LSTMs, be utilized to effectively integrate sentiment data with traditional market indicators for stock index prediction?

This question focuses on designing and optimizing a deep learning model that integrates both sentiment and traditional market data, exploring how LSTMs can capture complex temporal dependencies.

RQ5: To what extent does integrating sentiment data improve the performance of stock index prediction models compared to using quantitative data alone?

This question evaluates the added value of sentiment analysis by comparing the performance of the proposed model with traditional models, using metrics like RMSE, MAE, and R-squared to validate improvements.

RQ6: How should the sentiment of individual companies be weighted, considering that companies with larger market shares tend to dictate the overall value of the index?

This question explores methods for appropriately weighing the sentiment of each company in the DJIA, accounting for their market influence to ensure that sentiment analysis reflects the overall market dynamics accurately.

Structure Overview

1. **Literature Review:** This section examines existing approaches to stock market prediction, including traditional quantitative models and recent advancements in machine learning. It discusses how sentiment analysis is applied in finance, emphasizing methods for extracting and quantifying sentiment from financial news and social media. The review highlights research gaps, particularly the limited integration of multi-source sentiment data in index forecasting, forming the basis for this study's contributions.
2. **Methodology:** This chapter describes the research approach, including data collection through web scraping, sentiment analysis using VADER, and the design of an LSTM network to capture temporal patterns in the data

3. **Implementation:** This section outlines the application of the methodology, focusing on the tools and frameworks used, the development of the LSTM model, and strategies to address challenges in data preprocessing and web scraping.
4. **Results and Evaluation:** This chapter presents the findings of the study, including the performance metrics of the developed models. It reports on the predictive accuracy measured by RMSE, MAE, R-squared and compares the results of models with and without sentiment features. The chapter includes visualizations like loss curves and predicted versus actual price plots to illustrate model performance. It also assesses the model's robustness during periods of market volatility.
5. **Discussion:** This section interprets the results in the context of the research questions and the existing literature. It analyzes how the integration of sentiment analysis impacted the model's predictive performance and discusses the effectiveness of the data collection and feature engineering strategies. The discussion compares the study's findings with previous research, highlighting the unique contributions and differences. It addresses the limitations of the study and explores the implications of the findings for theory and practice.
6. **Conclusion and Future Work:** This final chapter summarizes the key findings of the research, emphasizing the impact of incorporating sentiment analysis on prediction accuracy for the DJIA index. It acknowledges the limitations of the current approach, such as reliance on news headlines and challenges with weighting individual company sentiment. The chapter provides recommendations for future research, including incorporating alternative sentiment sources, experimenting with advanced model architectures, and integrating additional data modalities to enhance predictive performance.

Literature Review

Stock Prediction Methods

Stock price prediction is a difficult task requiring scrutinizing a host of market factors. Over the years, the task's complexity has fostered the development of a host of techniques. Traditional stock price prediction techniques rely heavily on statistical techniques, such as time series analysis and econometric modeling. Approaches such as the Autoregressive Integrated Moving Average (ARIMA), Moving Average (MA), and Generalized Autoregressive Conditional Heteroskedasticity (GARCH) are widely established methods used to forecast changes in the stock market while modeling linear relations and capturing volatility in the market. As stated by Box & Jenkins (2015), these methods have given insight into simple market trends and have provided the foundation of financial modeling. While they have proved helpful in identifying trends and patterns, more often than not, they fail when applied to the nonlinear and complex behavior exhibited by financial markets, such as sudden spikes or drops initiated by investor sentiment or outside shocks. This limitation thus calls for more sophisticated methods to capture the detailed dynamics of the financial market.

The major shortcoming of these traditional techniques is their inability to model financial data's chaotic and nonlinear nature adequately. The movement of stock markets is determined by various factors, such as economic indicators, geopolitical events, and investor sentiment, which interact in complicated ways. Traditional statistical models include ARIMA and GARCH, in which assumed stability and linearity are quite foreign to real-world dynamics in the financial markets. Therefore, these models tend to be very poor when the market suddenly gets shocked or if there is an unexpected change in the market dynamics, such as in economic crises or unexpected political events. The failure to adapt to the rapidly changing conditions underlined the need for more sophisticated techniques that could effectively capture the diverse and interdependent factors influencing the markets. The research has opened doors to seek more advanced methods that could capture these more effectively.

The advent of machine learning brought another paradigm shift in methodologies for stock prediction, where evolving research interests became more biased toward models capable of handling big and complex datasets for discovering nonlinear relationships. Machine learning methods shine when it comes to uncovering intricate patterns in data—an essential attribute for financial forecasting because financial variables are chaotic and interdependent. Due to flexibility and capability for generalization across various data contexts, techniques like

Support Vector Machines, K-Nearest Neighbors, and decision trees have become prevalent methods for market prediction (Xiong et al., 2018). These models are instrumental in identifying patterns not visible by traditional statistical approaches and, thus, may be very useful for applications in stock market prediction.

More recently, deep learning methods have received much attention concerning the modeling of time-sensitive financial data for their efficiency in capturing temporal dependencies, which are often extremely important to understand when it comes to the forecasting of stock markets. Long Short-Term Memory Networks overcome the problem of vanishing gradients found in traditional RNNs, allowing the handling of long-term dependencies more effectively, which is an advantage in the conduction of analysis within the stock market (FinBERT-LSTM, 2022). Overcoming the flaws of RNNs, LSTMs finally offer the option of keeping information for longer, which is essential when trying to model prolonged sequences with lagging effects, so common in financial data.

By design, LSTMs have been made to keep information for a longer period, thus proving perfect for the long dependencies usually associated with the financial time series. This results from memory cells and gates that control the flow of information in and out of the cell, hence allowing LSTMs to decide at the end what information to retain or forget. This, in turn, enables LSTMs to learn from the fundamental pattern in the stock price movement over weeks or months rather than just short-term fluctuations. A possibility certainly exists that if any economic indicator influences the stock market over several months, the LSTM will capture that long-term relationship and hence result in better accuracy in predictions. As a result, LSTMs have been widely adopted in those areas and are among the most used deep learning models in financial forecasting.

Furthermore, hybrid models that incorporate LSTMs with other deep learning architectures have also been tried to enhance the prediction's robustness and accuracy. Combining LSTMs with CNNs, for example, allows the model to learn temporal and spatial features of financial data for deep insight into the driving factors in the process of stock pricing (Analytics Vidhya, 2021). These hybrid methods take into consideration the benefits of each type of model. While CNNs offer outstanding performance in pattern spatial identification, LSTMs catch temporal relationships; hence, combined, they assure better predictive performance. Recently, even attention mechanisms joined with LSTMs to help the model focus on the most relevant parts of the input sequence to further improve prediction accuracy.

The wide usage and combinations of LSTMs with other models into hybrid ones prove the significance of LSTMs in financial forecasting. Those especially fit the financial data's nonlinear, interdependent, and chaotic nature. This ability to learn long-term dependencies, added to their flexibility when combined with other architectures, makes LSTMs the cornerstone of modern financial modeling. As financial markets only keep developing and becoming more complex, so will the demand for sophisticated, adaptive models like LSTMs. This will equip investors and policymakers with superior tools to understand and predict market behaviors.

Sentiment Analysis in Finance

The integration of sentiment analysis into the models has become increasingly viewed as a promising way of improving the stock market's prediction accuracy. Sentiment analysis applies NLP techniques in opinion, emotion, and attitude extraction from textual data such as financial news, posts on social media, and analyst reports. Behind this approach lies the fact that market sentiment often leads to investor behavior. For example, negative news sentiment about the company or its industry can cause a sell-off and, consequently, a drop in stock price. (Explainable Stock Price Prediction, 2022).

A recent major milestone over the State-of-the-Art results in this domain was achieved by a domain adaptation of BERT called FinBERT, which was trained on financial data. In effect, FinBERT understands nuances of the language specific to finance, which is very helpful in accurate sentiment classifications of financial news articles and their ensuing impact on stock price movements. Indeed, various studies, such as Mittal et al. (2022), have shown the efficacy of integrating FinBERT with predictive models. Once the sentiment data is added to the traditional sets of quantitative features, the results indicate a significant improvement in forecast accuracy. Because of its capability to understand nuances associated with financial language, FinBERT is hugely effective at capturing sentiment data that directly influences investor decisions and market outcomes. Moreover, FinBERT outperforms standard NLP models when applied to financial data. It proves that domain-specific training is inevitable in the case of practical sentiment analysis within finance.

Moreover, the influence of social media on investor sentiment has become such a focal point that sites like Twitter and Reddit have been argued to be some of the top setters in shaping market perceptions. Social media platforms provide unimaginable data in real-time that can be used to estimate investor sentiment and predict market movements. Given the rising trend,

sentiment analysis tools are increasingly used to search these platforms for real-time changes in investor sentiments. For instance, a case would be that sudden spikes in negative sentiment, as detected on social media, correlate with market downturns and thus provide an early warning signal that could be integrated into the prediction models to make them responsive. Extracting real-time sentiment from social media becomes vital for developing adaptive models that can promptly respond to changes in market sentiment, especially at heightened periods of market activities. This real-time sentiment analysis will make predictions dynamic over market activities and serve investors with better decision-making abilities after increased and unforeseen turns in public opinion.

Integrating financial news and social media sentiment data into predictive models provides detailed insights into market dynamics. That is because, as opposed to financial news, which may delve into deep analyses and include expert opinion, social media provides uncooked, raw sentiment across various investors and market participants. These two sources of sentiment combined help get a more holistic view of the market condition. When news reports are also favorable, a spike in positive sentiment related to a stock on Twitter may indicate a substantial buying opportunity. Conversely, any divergence between the sentiments expressed in news reports and those reflected on social media could signal imminent risks or market overreactions that one may want to avoid.

Other important aspects include temporal features of sentiment. Sentiment is not a constant entity; it changes how far its effect would reach stock prices over time. Another recent improvement in sentiment analysis includes temporal features, such as tracking the change in sentiment over several days or weeks to understand its implications on market movements. Analyzing such temporal patterns will make predictive models robust, capturing immediate and lagged responses to sentiment changes. This will be important for financial markets since the timing of information dissemination and investor reaction can significantly affect price movements.

Sentiment analysis remains an ongoing development in finance, with newer methods often devised to improve accuracy and applicability. The sophistication of text interpretation and the extraction of actionable insights from such complex financial texts is directly predicated on improving NLP advances such as transformer-based models and attention mechanisms. These will pave the way for more sophisticated models integrating sentiment data with other

financial indicators, leading to better predictability performance and, therefore, more informed investment strategies.

Research Gaps and Challenges

Although much work has recently been done in integrating sentiment analysis with machine learning models for stock prediction, many gaps and challenges still need to be met. Among these issues is how to effectively integrate a stock index's sentiment data, which involves aggregating the sentiments of many companies comprising a particular stock index. Each of these companies influences the market differently, and treating all sentiments of the exact nature will lead to wrong conclusions. This would involve the development of an appropriate mechanism for aggregating and weighting sentiment based on the company's market share and how it would influence the movement of the overall index. For instance, larger companies within the DJIA may need to have their sentiment weighed more in relation to the smaller components since they usually have a greater influence on the eventual index. This significant complexity layer still needs to be captured in research into modeling movements in stock index prices.

Second, data aggregation is challenging, especially in keeping data consistent when processing multi-source sentiment data. The quality, timeliness, and perspective of financial news sources vary so frequently that it is very difficult to aggregate these variant sources into one measure of unified sentiment. Weighting the sentiment data according to source reliability, credibility, and context of the news will be crucial to constructing a sound sentiment analysis framework. While such issues have been identified in the prevailing literature of Saxena (2023), comprehensive solutions to handle the heterogeneity of multi-sourced sentiment data and their reliability still need to be revised.

Moreover, the role of sentiment regarding stock index movement during high market volatility has yet to be widely researched. The magnitude of sentiment varies with the state of the market, and constructing models that adapt to it is quite a task. Market sentiment is usually more volatile during periods of economic uncertainty or geopolitical tension, and traditional models cannot adapt to these changes fast enough. Another critical research gap exists in the development of adaptive models, those that alter their behavior given changes in the prevailing market conditions.

Most of the literature currently on sentiment-driven stock price prediction is for individual stocks, not for stock indices. Methods of predicting individual stock prices based on

sentiment cannot be applied to the prediction of stock indices since aggregating sentiment from many companies, each with a varying influence on market performance, is needed for the latter. Robust models should be developed to predict stock index prices, as these indices are comprehensive indicators of market performance, portraying insights that critical areas are supposed to guide investors and policymakers in making well-informed decisions.

Aggregation at an index level presents different challenges, as it needs to capture the integrated impact of the sentiment of all the constituent companies and a share of the market. However, the approach towards aggregating sentiment on an index would differ in terms of the effects of accounting sentiment on constituent companies and with different shares in the market. Developing robust models for predicting stock index prices is crucial because the representations provided by the indices will be greater about the overall market performance, thus making them beneficial means for investors and policymakers. Current methods typically use simple averaging or weighting of individual sentiments without fully capturing the dynamic nature of each company's sentiment influencing the index. Consequently, there is great potential for improving the accuracy of index-level predictions by developing a more sophisticated weighting mechanism to dynamically adjust sentiment with market events, company performance, and the broader economic environment. It also has to account for possible spillover effects across firms, where news about one company could impact investor sentiment regarding other firms in that particular industry or sector.

Besides, there is also a growing demand to include temporal aspects of sentiment analysis in a more subtle manner. Most current models use static sentiment features and need to consider the temporal dynamics of change in sentiments fully. Incorporation of the temporal aspect-lagged values of sentiments and sentimental moving averages can help capture the lagged effects of news on stock prices. Understanding changes in sentiment over time and how such changes affect investor behavior may suggest a deeper understanding of the dynamics of the market-specifically for those indices that represent several companies, each with their respective influences on overall market trends.

Methodology

Data Sources and Collection

This research relies on financial and sentiment data for the Dow Jones Industrial Average (DJIA) and its component companies, focusing on the period from 2022 to 2023. This timeframe was deliberately selected to explore market behavior during the post-pandemic recovery and episodes of heightened volatility. By covering this dynamic period, the study examines how markets react to significant global and regional news events. Key financial indicators, including daily open, close, high, low, and trading volume, were gathered through the Yahoo Finance Python library, ensuring reliable and timely data inputs for the predictive model.

For sentiment analysis, financial news headlines were sourced from reputable platforms, including Markets Insider, which aggregates content from various outlets such as CoinDesk, Seeking Alpha, TipRanks, and RTTNews. Additionally, headlines from the Financial Times were incorporated to ensure a comprehensive dataset. The scraping process utilized Selenium and BeautifulSoup for efficient data extraction. Selenium was particularly effective for navigating and handling dynamic web pages that required JavaScript rendering, which is common on modern financial websites. Anti-scraping measures such as random delays, headless browsing, and the use of custom user agents were implemented to avoid detection and ensure uninterrupted scraping. BeautifulSoup was subsequently employed to parse and extract specific content from the HTML structure of the pages, such as news headlines, publication dates, and article links. By combining these scraping techniques and using multiple news sources, the study aimed to provide a comprehensive sentiment profile of the market, reducing potential biases that might arise from relying on a single news outlet. This approach not only ensured a diverse dataset but also enabled a more robust sentiment analysis that better reflected the true market conditions.

Sentiment Analysis Techniques

To quantify sentiment from the collected news data, the study employed VADER (Valence Aware Dictionary and Sentiment Reasoner). VADER was selected for its effectiveness in analyzing short texts, such as news headlines, by assigning compound sentiment scores that indicate positive, neutral, or negative sentiment. One of the major strengths of VADER is its ability to capture the intensity of emotions in text, making it particularly well-suited for

financial news where sentiment can fluctuate significantly. The tool utilizes a pre-built lexicon that includes words and phrases commonly used in everyday language, assigning weights to them to calculate the overall sentiment score. This approach allowed for the accurate assessment of sentiment polarity in the context of market news, providing insights into whether news headlines were likely to drive optimism or pessimism among investors.

VADER was particularly useful for providing a general sentiment overview of daily market news by simplifying the sentiment into easily interpretable scores. These sentiment scores were critical in identifying short-term market mood shifts and understanding broader trends across different time frames. Moreover, VADER's ability to efficiently handle large volumes of news data made it an ideal choice for this study, where headlines from multiple sources were processed. This tool provided a reliable method to gauge market sentiment on a daily basis, forming the basis for the sentiment features engineered later in the modeling process.

Feature Engineering

Feature engineering was a critical aspect of the study, focusing on transforming raw sentiment and financial data into features that could be effectively utilized by the predictive model. Aggregated sentiment data from the multiple companies in the DJIA was used to create overall market sentiment features. One of the key features developed was volume-weighted sentiment, which was calculated by weighting each company's sentiment score according to its trading volume. This ensured that companies with a larger trading volume, and thus a larger influence on the DJIA, had a proportionate impact on the aggregated sentiment feature. By incorporating this volume-weighted approach, the model was better equipped to reflect the influence of major market players whose trading activities significantly affect the index.

To further capture the effects of sentiment on stock prices over time, several temporal features were engineered. Lagged sentiment values (e.g., 1-day, 2-day, 3-day lags) were introduced to model the delayed market reaction to news events, helping the model understand how past sentiment influences current price movements. These lagged features allowed the model to capture the inertia in market reactions, where sentiment from previous days could continue to impact trading behavior and price movements.

Exponentially Weighted Moving Averages (EWMA) of sentiment scores were calculated to smooth short-term sentiment fluctuations while highlighting longer-term sentiment trends. The EWMA provided a balanced view by reducing the noise from daily sentiment changes

and emphasizing sustained trends that are more likely to influence investor behavior. This was particularly important in distinguishing between transient news impacts and more persistent market sentiment shifts.

Additional features, such as sentiment volatility (measured as the rolling standard deviation of sentiment scores) and sentiment-to-volatility ratios, were also generated to capture the dynamic nature of market sentiment. Sentiment volatility was designed to measure the variability in sentiment over a specific window, offering insights into periods of heightened uncertainty or strong market reactions to news. The sentiment-to-volatility ratio further helped in understanding the proportional impact of sentiment changes relative to their variability, providing a nuanced metric to gauge the stability of sentiment trends.

These engineered features ensured that both immediate and delayed impacts of news sentiment on stock prices were represented, enhancing the model's ability to predict market movements effectively. By incorporating these advanced sentiment features, the model was able to better capture the complex interactions between news sentiment, market perception, and price behavior, ultimately leading to improved prediction accuracy and a more comprehensive understanding of the factors influencing the DJIA.

Exploratory Data Analysis (EDA)

EDA was performed to analyze the insights provided by the engineered sentiment features and their relationship with the DJIA stock prices. Correlation heatmaps were used to visually assess the connections between sentiment features and DJIA closing prices, offering a clear view of how sentiment influences stock price movements. These visualizations underscored the importance of including sentiment data in the model, as they demonstrated improved predictive performance compared to models that excluded sentiment inputs. The EDA was instrumental in identifying the most impactful features for predicting stock price movements, such as lagged sentiment values and sentiment volatility, which emerged as critical components. By incorporating these advanced sentiment features, the model could better capture the intricate relationships between news sentiment, market perceptions, and stock price dynamics, ultimately enhancing prediction accuracy and providing deeper insights into the factors affecting the DJIA.

Model Development and Evaluation

This study used Long Short-Term Memory (LSTM) networks to predict the DJIA index price, chosen for their ability to process sequential data and identify temporal patterns effectively. Their capacity to retain critical information over time makes them highly suitable for financial market analysis, where trends and dependencies span multiple time steps. TensorFlow, a reliable and scalable deep learning framework, was employed to design and implement the model. The model combined traditional financial features—such as open, high, low, and close prices—with sentiment-based features derived from news headlines, allowing for a more holistic integration of numerical and qualitative insights. To ensure optimal performance, hyperparameter tuning was carried out using grid search, systematically evaluating combinations of LSTM units, learning rates, batch sizes, and training epochs. This approach enabled the identification of the most effective configuration for the predictive task. The dataset was divided into training (80%) and testing (20%) splits, providing a robust method to assess the model's ability to generalize to unseen data while reducing the risk of overfitting.

To evaluate the model's accuracy, key metrics such as RMSE, MASE, and R^2 were used. RMSE was prioritized due to its sensitivity to significant errors, making it ideal for financial predictions where minimizing large deviations is critical. MASE offered a scale-independent accuracy measure, enabling meaningful comparisons across datasets, while R^2 quantified the proportion of variance in stock prices explained by the model, highlighting its predictive power. A baseline model without sentiment features was also used for comparison, demonstrating that including sentiment data substantially improved prediction accuracy and emphasized the role of market sentiment in shaping stock prices. To further validate the model, training and validation losses were closely monitored across epochs to ensure effective learning without overfitting. Early stopping was applied to halt training when validation loss stopped improving, maintaining a balance between model complexity and generalization. Additional experiments examined the sensitivity of the model to variations in hyperparameters, offering insights into its robustness and stability. By integrating sentiment features with traditional financial metrics, the study developed a predictive framework capable of capturing the nuanced interactions between market sentiment and stock prices. Comparing RMSE values, feature importance, and training behaviors revealed the significant advantage of combining qualitative sentiment data with quantitative financial data, leading to improved accuracy and deeper insights into stock market dynamics.

Implementation and Tools

Data Collection and Pre-processing

The integration of data sources was a critical step in the implementation process, combining financial and sentiment data from multiple origins. Financial data, including daily open, close, high, low, and trading volume for the Dow Jones Industrial Average (DJIA), was obtained using the Yahoo Finance Python library. This data provided essential metrics for analyzing stock movements, offering insights into price fluctuations and trading volume, which were later combined with sentiment analysis.

Sentiment data was collected by web scraping news articles from Market Insider and Financial Times, two reputable sources known for delivering timely financial updates. Articles from Market Insider were retrieved using a combination of the requests library and BeautifulSoup for HTML parsing. The process involved generating dynamic URLs for Dow Jones constituent companies, fetching the HTML content, and extracting relevant details such as headlines, publication dates, sources, and links. These extracted articles were stored in a Pandas DataFrame, with multiple pages scraped per company to ensure a comprehensive dataset spanning both recent and historical events.

For Financial Times articles, a more advanced scraping technique was required due to the dynamic nature of the website, which relies heavily on JavaScript rendering. Selenium was employed to simulate browser interactions, allowing the retrieval of dynamic content. The process included navigating through search results for each company, scrolling through pages, and waiting for content to load. Once the HTML was rendered, BeautifulSoup was used to parse the headlines, publication dates, and URLs. Anti-scraping measures, such as random delays between requests, headless browser mode, and custom user-agent strings, were implemented to avoid detection and mitigate the risk of IP blocking. This two-step method ensured a rich and diverse dataset of articles, which served as the foundation for deriving sentiment indicators related to the DJIA.

Incorporating Sentiment into Data

After extracting sentiment scores, these were merged with the financial data to create a unified dataset. Both datasets were synchronized by aligning their date formats, ensuring consistency across records. Sentiment data was aggregated at the daily level, resulting in a

unified metric that matched the granularity of the financial data. This integration established the groundwork for subsequent feature engineering and model development.

Data Cleaning

Once the data was gathered, a rigorous cleaning and normalization process was performed to ensure its reliability and consistency. Duplicate articles were identified and removed by analyzing the "title" field, eliminating thousands of redundant entries. This step was crucial to prevent biased sentiment analysis caused by overrepresentation of certain articles.

Sentiment Analysis Deployment

VADER was employed to analyze the sentiment of the news headlines. It is a lexicon and rule-based tool specifically intended for limited-length text such as headlines. The ability to ascertain subtle sentiment even within small pieces of context-rich data drove this tool's choice. You first need to initialize the `SentimentIntensityAnalyzer` from the VADER library, which will be used to execute the scoring. Cleaner processing was done using a custom function, `get_sentiment_score()`. This function wraps VADER's `polarity_scores()` method, which generates four scores: positive, negative, and neutral, besides the compound. The compound score, ranging from -1 to 1, represents the most negative and the most positive sentiment, respectively, and it was picked for this analysis because it provides a simple, overall measure of sentiment. These scores are generated based on VADER's pre-built dictionary of sentiment-laden terms and rules for adding context.

This function was applied efficiently to each headline using Pandas's `apply()` function. In this manner, it computed sentiment values for all headlines across a new column in the `DataFrame`. It prepared this dataset for further statistical analysis and modeling by quantizing qualitative text into numerical sentiment scores. Further refinement was done to the sentiment data to integrate well with other datasets. The date column in the `DataFrame` was then converted into a `DateTime` object using Pandas' `to_datetime()` function since the recording frequency of the sentiment data is at headline levels. Typically, this is done because recordings of various other datasets are usually done daily. Thus, having the same date format allows the sentiment scores to be matched with financial and temporal data to enable meaningful analysis from each base. This was quite a practical and reliable approach using VADER to transform unstructured text into meaningful numerical scores for sentiment analysis. The implementation was carried out efficiently by leveraging the `SentimentIntensityAnalyzer` and its scoring methods. This would give metrics that were

accurate and relatively easy to combine with financial data, hence upholding a good framework for extended analysis.

Feature Engineering

Volume-Weighted Sentiment Score: This feature was designed to consider the difference influence of stocks on the index. The sentiment score of each company's stock was multiplied by the trading volume of respective stock. Across all stocks, the product was normalized. This ensured that companies with more trading activity had more impact on the Dow Jones sentiment metric. In other words, firms that had more trading volume had more impact on this sentiment. Assuming the sentiment score of Apple is 0.9 and the trading volume is 5 million and if the sentiment score of Microsoft is 0.7 and the trading volume is 3 million, their contribution to the volume-weighted sentiment score will be as follows. Apple will give $0.9 \times 5,000,000 = 4,500,000$ and Microsoft will give $0.7 \times 3,000,000 = 2,100,000$. When we fired everything up the total was 6.6 million. When we divide this amount by the total volume traded worth 8,000,000 ($5,000,000 + 3,000,000$), the score translates into 0.825 of a volume-weighted sentiment score. This result reflects the overall sentiment placing greater weight on Apple given its larger trading volume.

$$VW_Sentiment_Score = \frac{\sum (Sentiment_Score \cdot Trading_Volume)}{\sum Trading_Volume}$$

Fig 1

Exponentially Weighted Moving Averages (EWMA): To better capture sentiment trends, exponentially weighted moving averages of sentiment scores were computed over various time spans (3, 7, 14, and 30 days). The EWMA helped smooth out daily fluctuations in sentiment and emphasized more recent observations, making it useful for tracking the momentum of market sentiment over time. The different spans allowed the model to consider both short-term and long-term sentiment trends, which could impact stock movements in different ways.

Cumulative Sentiment Change: This feature was created by calculating the difference between the 7-day EWMA and the 3-day EWMA of sentiment scores. This metric was designed to capture recent shifts in sentiment trends, providing insights into whether market sentiment was becoming more positive or negative compared to recent averages. Cumulative sentiment change is particularly useful for detecting abrupt changes in sentiment that could trigger significant market reactions.

Sentiment Volatility: Sentiment volatility was calculated as the rolling standard deviation of daily sentiment scores over a 5-day window. This feature captured the variability of sentiment, providing an indication of how stable or unstable market sentiment was over a given period. Higher sentiment volatility could indicate uncertainty or mixed opinions in the market, which might correlate with higher stock price volatility.

Lagged Sentiment Features: Lagged sentiment features were introduced to model the delayed effect of sentiment on stock prices. Lagged values of the sentiment score (1-day, 2-day, and 3-day lags) were included to capture the influence of past sentiment on current market behavior. This approach helped the model understand temporal dependencies and provided insights into how past sentiment influenced subsequent price changes.

Sentiment-to-Volatility Ratio: This feature was calculated as the ratio between the 7-day EWMA of sentiment scores and the sentiment volatility. It provided a measure of how significant the sentiment signal was compared to its variability. A higher ratio indicated strong sentiment with relatively low volatility, which could be indicative of clearer market direction.

Volume-Weighted Sentiment Feature: In addition to the overall sentiment score, a volume-weighted sentiment feature was calculated by multiplying the sentiment score by the daily trading volume for each stock. This feature aimed to capture how significant the sentiment was relative to the trading activity, reflecting the intensity of market reactions to news based on trading volume.

Finally, the dataset was cleaned by handling any NaN values that emerged during the feature engineering process. Missing values were filled with 0 where appropriate, and the dataset was checked for consistency to ensure it was ready for training.

Exploratory Data Analysis

Distribution of Dow Jones Closing Prices (2022-2023)

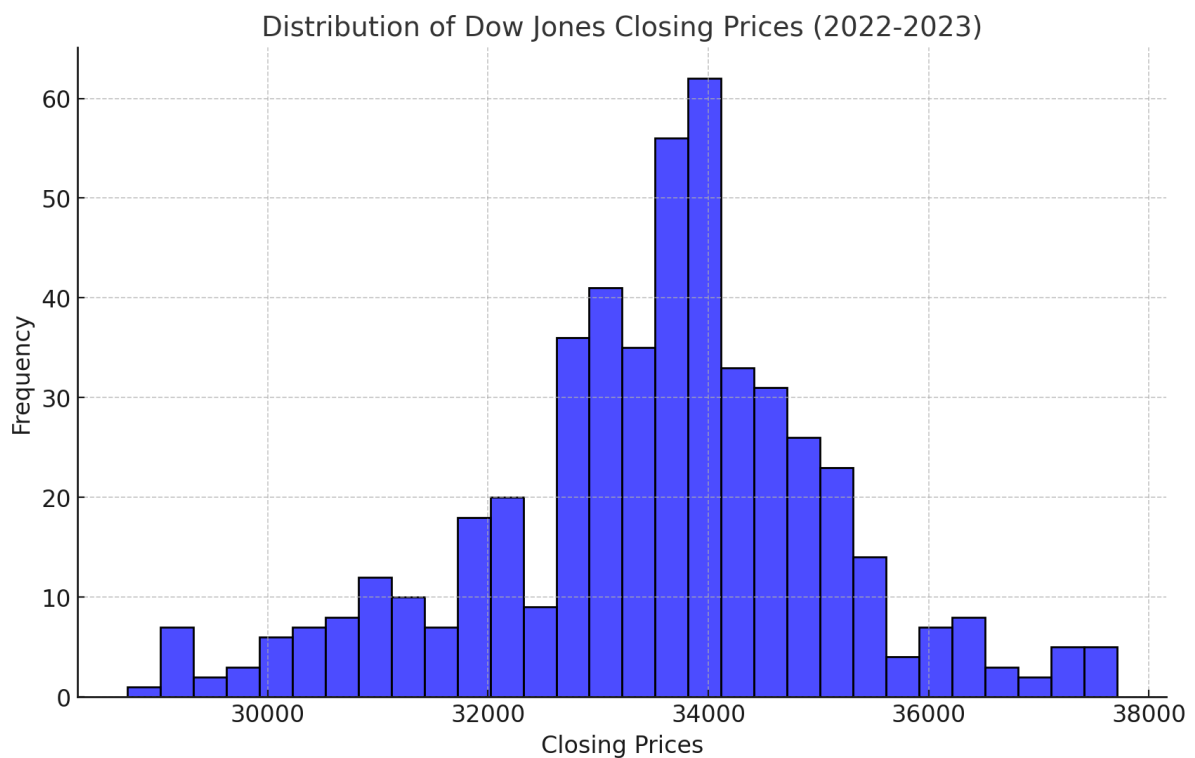


Fig 2

The histogram of Dow Jones Industrial Average (DJIA) closing prices (Fig 2) for 2022–2023 shows a clear clustering within a central range, reflecting the index's relative stability despite economic challenges such as inflation, interest rate hikes, and geopolitical events. The limited variability in extreme highs or lows underscores the resilience of the Dow, with no prolonged dominance of bullish or bearish trends.

This distribution is particularly relevant to the context of sentiment analysis in predicting stock prices. The consistency in the closing price range suggests that external factors, including market sentiment, played a crucial role in maintaining this stability. By correlating sentiment features with this distribution, we can explore whether sentiment trends contributed to the observed clustering or if specific sentiment-driven anomalies align with periods of deviation. This insight aligns with the broader goal of analyzing the Dow's price behavior to understand the influence of news sentiment on market dynamics.

Correlation Matrix Heatmap

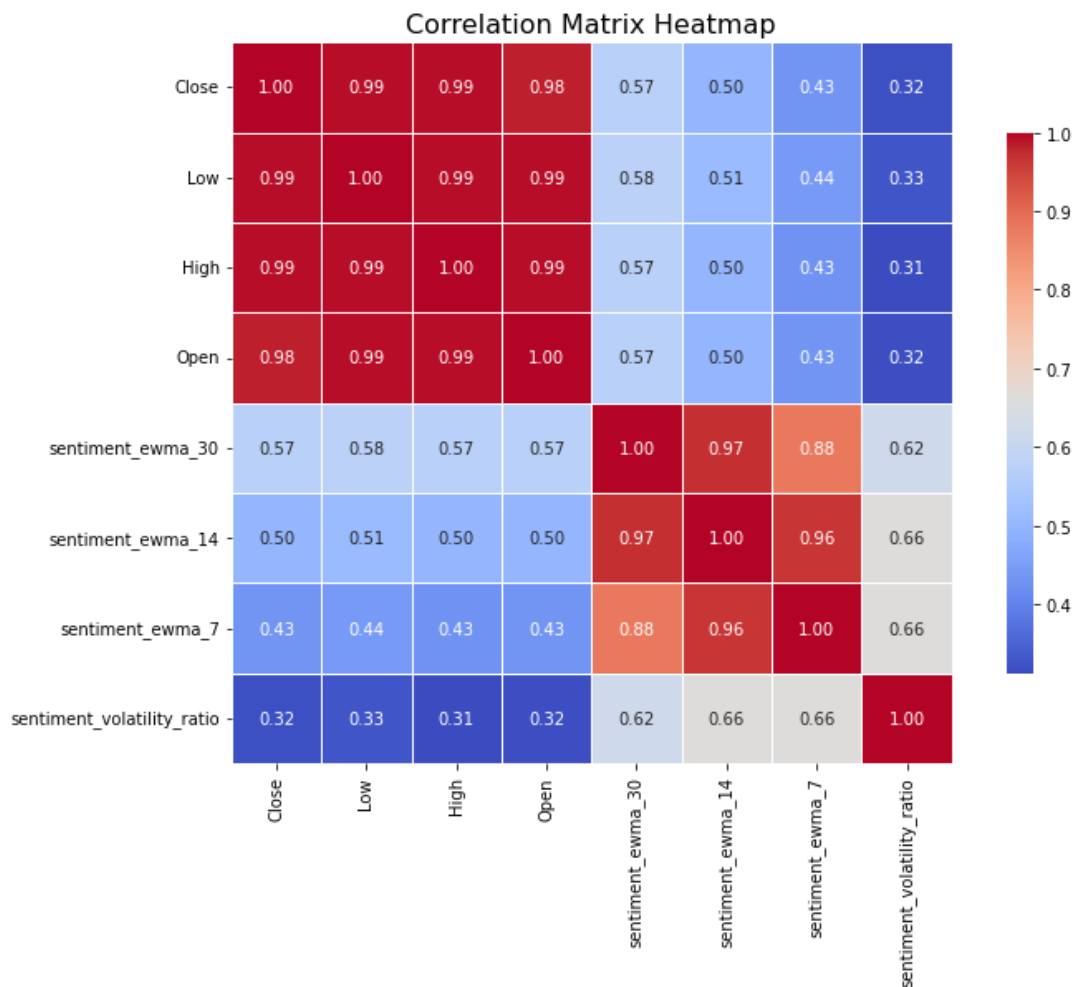


Fig 3

The correlation heatmap (Fig 3) provides a visual representation of the relationships between various sentiment features, all of which originate from the volume-weighted sentiment score. This foundational metric was calculated by multiplying each company's sentiment score by its respective trading volume, ensuring that companies with higher trading activity exert a proportionate influence on the aggregated sentiment for the Dow Jones index. This approach reflects the true market impact of sentiment by emphasizing the role of highly traded stocks in shaping overall sentiment trends. Every derived sentiment feature—such as exponentially weighted moving averages (EWMA), cumulative sentiment change, sentiment volatility, lagged sentiment features, and the sentiment-to-volatility ratio—ultimately builds on this volume-weighted sentiment metric, offering a nuanced understanding of market behavior.

From the heatmap, several notable correlations emerge. The short-term EWMA (e.g., 3-day and 7-day) exhibit an exceptionally strong positive correlation with each other, highlighting their similarity in capturing recent sentiment trends. However, their correlation slightly

decreases when compared to the longer-term EWMA (14-day or 30-day), illustrating the difference in how short-term and long-term sentiment patterns evolve. By grounding all these features in the volume-weighted sentiment score, the framework ensures a consistent representation of market sentiment influenced by trading activity. This ensures that every derived feature retains its connection to the original metric, capturing both the intensity and the stability of market reactions to sentiment changes, which can then inform predictive models and market analysis.

The volume-weighted sentiment feature's high correlation implies that it captures both the intensity of market reactions (through trading volume) and sentiment direction, making it a strong candidate for enhancing predictive accuracy.

The EWMA features provide a smoothed version of sentiment trends, which is beneficial for capturing the overall momentum of sentiment rather than being affected by short-term noise. Including multiple EWMA with different time frames allows the model to capture both short-term shifts and longer-term sentiment trends.

Sentiment EWMA 30 Vs Closing Prices

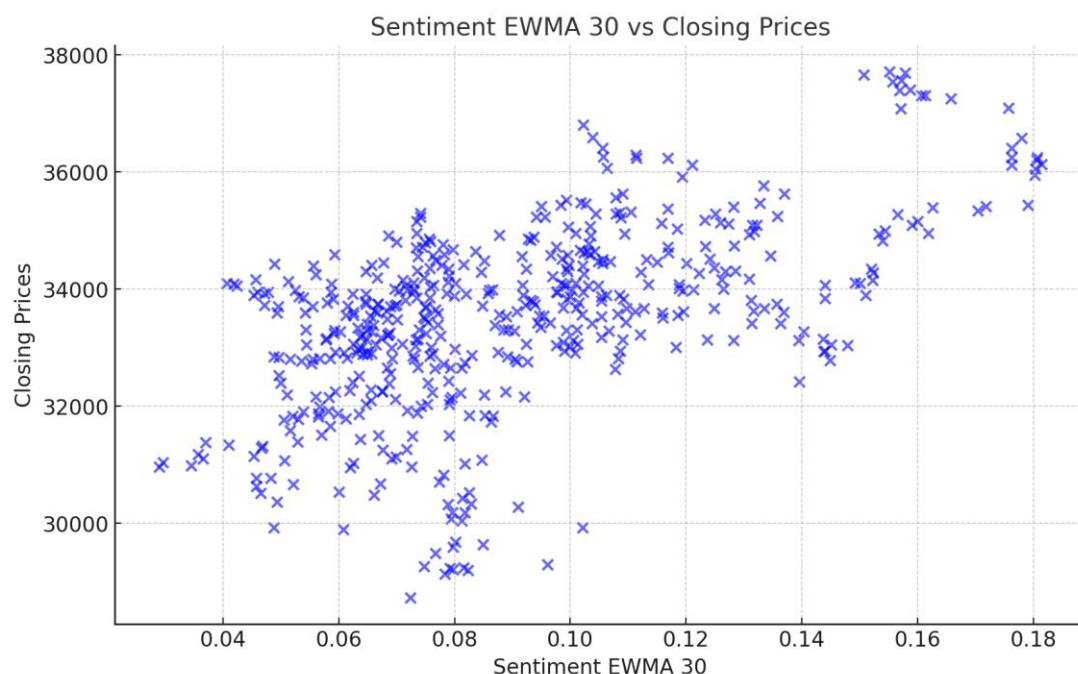


Fig 4

The Sentiment EWMA 30 (Exponential Weighted Moving Average with a 30-day window) (Fig 4) represents long-term sentiment trends by smoothing daily sentiment fluctuations over a rolling 30-day period. This metric prioritizes recent sentiment values while maintaining historical context, making it ideal for identifying sustained trends in market sentiment. In the scatterplot, Sentiment EWMA 30 shows a distinct positive relationship with closing prices. Higher EWMA 30 values are consistently associated with higher closing prices, with tighter clustering observed compared to short-term sentiment measures. This indicates that long-term sentiment trends are more reliable and less influenced by transient market volatility.

Notably, at lower sentiment levels, the variability in closing prices is minimal, further emphasizing the stabilizing effect of long-term sentiment. Even during periods of reduced sentiment, the Dow Jones index maintains relative stability, reinforcing the robustness of EWMA 30 as an indicator. The strong clustering and clear relationship in this plot highlight Sentiment EWMA 30 as a crucial feature for predicting long-term price movements. Its ability to capture sustained sentiment trends provides a reliable foundation for integrating sentiment analysis into predictive models.

Sentiment Volatility Ratio Vs Sentiment EWMA 30

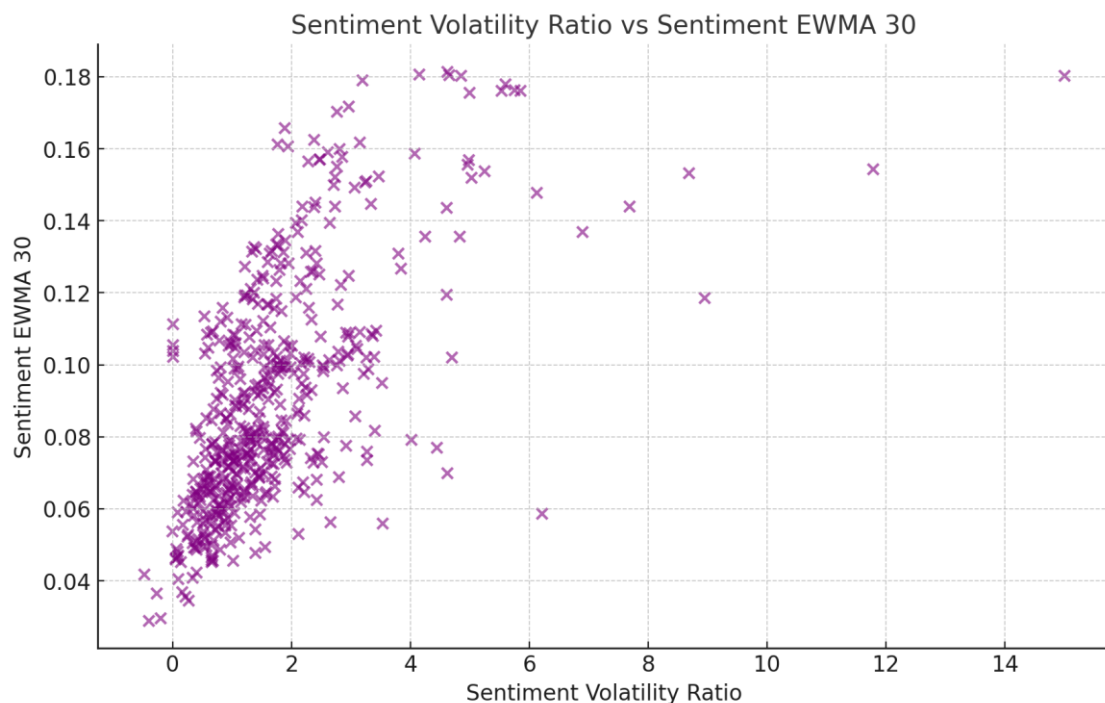


Fig 5

The (Fig 5) scatterplot illustrates the relationship between the Sentiment Volatility Ratio (a measure of how much sentiment fluctuates) and the Sentiment EWMA 30 (a long-term sentiment trend). With majority of data points are concentrated in the lower volatility range, reflecting that market stability is a dominant condition within the dataset. However, there are distinct outliers where higher volatility is observed, often aligning with medium-range EWMA values. This suggests that during periods of moderate long-term sentiment, market sentiment tends to fluctuate more. Conversely, when EWMA values are either strongly positive or negative, volatility decreases, indicating that sentiment stabilizes in strongly bullish or bearish markets. High sentiment volatility thus acts as a potential signal for market uncertainty or critical turning points, particularly when long-term sentiment trends are neutral.

Model Development and Training

Configuring the LSTM Model:

The LSTM model was built using sequential architecture model. The architecture utilized a single LSTM layer with 128 memory cells, followed by a Dropout layer to mitigate overfitting. The LSTM layer was set up without return sequences, meaning that all we needed in the output was a single value, the next closing price prediction. A Dense layer of one unit was attached at the end to create output value. The model was compiled with the Adam optimizer and mean squared error (MSE) loss function, which is typically used for regression to minimize the error in prediction.

The model configuration strongly relied on hyperparameter tuning. After trying out values, the number of LSTM units was finally set to 128 so as to not overload the model complexity. To prevent overfitting, a rate of 0.3 for dropout was selected while the batch size was set up as 16. The early stopping patience was set for 20 epochs which means that training will stop if there has been no improvement in the validation loss for 20 times.

Training Process:

The dataset was divided into an 80:20 ratio for training and testing purposes. The training data was scaled using the MinMaxScaler to normalize feature values between 0 and 1, ensuring the model could train efficiently without being biased towards features with larger ranges. The same scaling was applied to the testing set. A look-back period of 60 timesteps

was used, meaning the model used the previous 60 days of data to predict the next day's closing price. This choice aimed to capture short- and medium-term trends effectively.

During training, the model was fitted using a batch size of 16 over 50 epochs, with early stopping and model checkpointing to save the best-performing version of the model based on validation loss. The use of early stopping ensured that the model did not overfit to the training data, while model checkpointing helped preserve the best weights observed during training.

Testing and Evaluation

Implementation of Testing Protocols

The testing protocol involved splitting the data into training and testing subsets to validate model performance on unseen data. The trained model was evaluated on the test dataset using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), R-squared (R^2 Score) as the primary metric. The test dataset included data points that were scaled using the previously fitted MinMaxScaler to maintain consistency with the training process. The predictions on the testing dataset were transformed back to the original scale to facilitate direct comparison with actual values.

Results and Analysis

LSTM Without Sentiment Features

Performance Analysis

The LSTM model was initially trained without incorporating any sentiment-based features which are 'Close', 'Low', 'High', 'Open'. The model's performance throughout the training process is summarized by the epoch-wise validation loss and training loss values. Below are the notable details observed during the training process:

Epoch 1/50: The training loss started at 0.0586, while the validation loss improved to 0.03442. This initial high value indicated that the model required a few epochs to stabilize and begin learning the key patterns in the data.

Epoch 5/50: The validation loss further improved to 0.00587, demonstrating that the model was gradually converging and effectively capturing important relationships within the data.

Continuous monitoring of the validation loss was essential to determine the best weights and mitigate any overfitting risks.

Epoch 11/50: The validation loss achieved its lowest value of 0.00175. At this point, early stopping was triggered to prevent overfitting, ensuring that the model maintained optimal validation performance without further degradation. This mechanism was crucial in saving computational time and resources by halting training once the optimal performance was reached.

The loss plot (Fig 6) for the LSTM without sentiment features showed a steady decrease in training loss, converging well over time. The validation loss exhibited significant improvement early on, stabilizing around epoch 11. However, beyond this point, validation loss began to fluctuate, indicating challenges in generalizing to unseen data—likely due to the lack of additional contextual information from sentiment features.

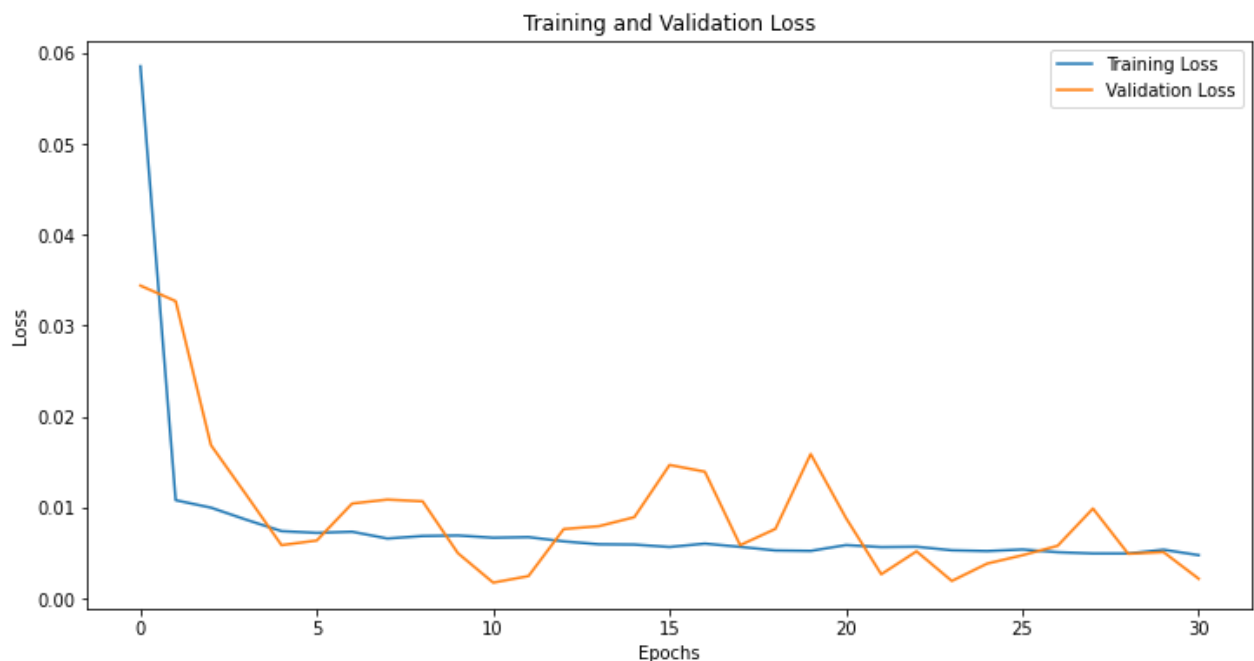


Fig 6

The predicted vs. actual plot (Fig 7) revealed that the model could capture the general upward trend of closing prices but struggled with precision, especially in capturing sudden peaks and troughs. These discrepancies highlighted the model's difficulty in fully

understanding market volatility without sentiment data. During periods of high volatility, the model tended to lag or miss significant price movements, underscoring the limitations of relying solely on numerical features.

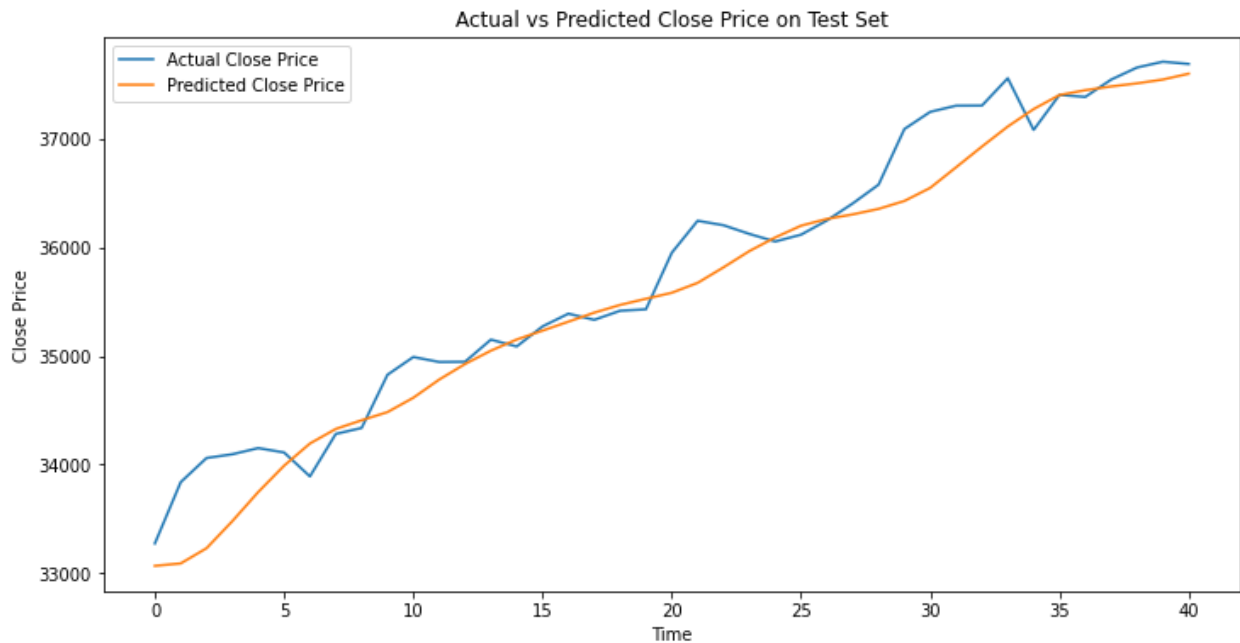


Fig 7

LSTM With Sentiment Features

Performance Analysis

The LSTM model was trained with the inclusion of sentiment-based features which are 'Close', 'Low', 'High', 'Open', 'sentiment_ewma_30', 'sentiment_ewma_14', 'sentiment_ewma_7', 'sentiment_volatility_ratio', Derived from financial news headlines and integrated with the model inputs. By incorporating these features, the model aimed to capture investor sentiment, a critical driver of market behavior. The training and validation loss progression for the LSTM model with sentiment features is outlined below:

Epoch 1/50: The training loss started at 0.0516, with validation loss improving to 0.00444. Similar to the previous model, these initial values indicated that the model required a few epochs to stabilize. Notably, the initial decrease in loss was slightly more rapid compared to

the model without sentiment features, suggesting that the sentiment data provided a richer context from the outset.

Epoch 5/50: The validation loss improved significantly to 0.00153, indicating that the sentiment features added valuable information for capturing market behavior. This improvement hinted at the model's ability to learn the emotional drivers behind price movements, which are often not captured by purely numerical features.

Epoch 41/50: The validation loss achieved its lowest value of 0.00099. At this point, early stopping was triggered to prevent overfitting, ensuring that the model maintained the best possible validation performance without further degradation. The early stopping mechanism was pivotal in saving computational time and resources by avoiding unnecessary further training once the optimal performance was achieved.

Throughout the training, early stopping and checkpointing were utilized to preserve the best model based on validation loss. The inclusion of sentiment features enhanced the model's ability to understand market trends, as evidenced by the lower validation losses compared to the model without sentiment features. This improvement was particularly notable in the model's ability to reduce overfitting, suggesting that sentiment data contributed to making more generalized predictions.

The loss plot (Fig 8) for the LSTM with sentiment features demonstrated a more stable and faster convergence compared to the model without sentiment features. The validation loss decreased more consistently, reaching its minimum at epoch 41 and stabilizing thereafter. This behavior indicates that sentiment data helped the model better generalize and adapt to the validation set, reducing overfitting and improving model stability. The training process was marked by less variability in loss, implying that sentiment information provided a smoother learning process by offering an additional layer of understanding of market dynamics.

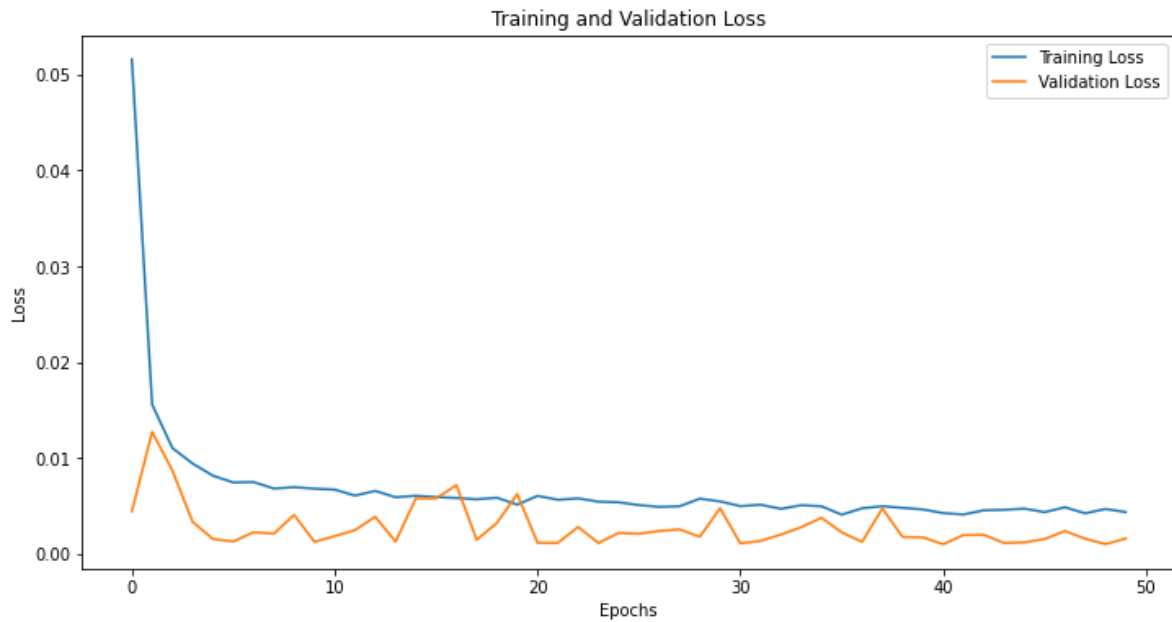


Fig 8

The predicted vs. actual plot (Fig 9) showed that the LSTM with sentiment features had a much closer alignment with the actual closing prices, especially during periods of high volatility. The model accurately captured smaller price fluctuations, reflecting an enhanced understanding of market sentiment and its effects on stock prices. By incorporating sentiment data, the model exhibited a greater ability to adjust predictions in line with unexpected market shifts, which were likely influenced by news and investor sentiment.

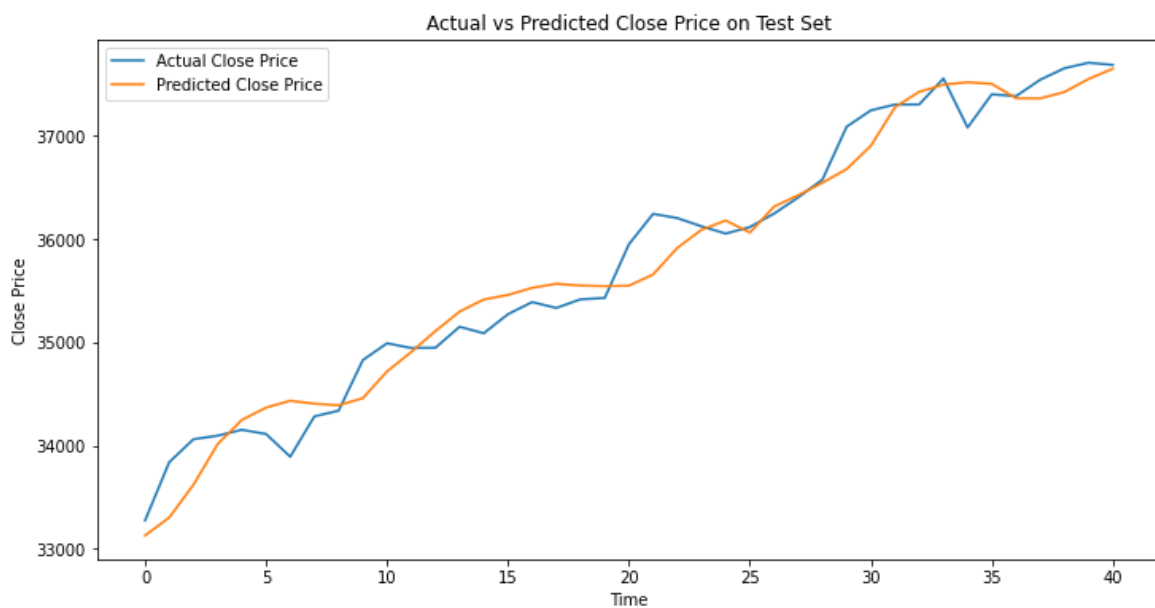


Fig 9

Comparison Of Models

	Metric	Without Sentiment Features	With Sentiment Features
0	Train RMSE	614.20	454.12
1	Test RMSE	337.48	253.78
2	Train MAE	462.72	352.41
3	Test MAE	247.70	198.47
4	Train R ²	0.81	0.90
5	Test R ²	0.93	0.96

Fig 10

The comparison between the LSTM models (Fig 10) with and without sentiment features underscores the substantial advantages of integrating sentiment data into the predictive framework. This integration significantly enhances various aspects of model performance, including training and validation loss, Root Mean Square Error (RMSE), Mean Absolute Error (MAE), R-squared (R^2), predictive accuracy, overfitting mitigation, and computational efficiency.

Initially, both models demonstrated different behaviors in their training and validation loss trajectories. The LSTM model without sentiment features started with a training loss of 0.0586 and a validation loss of 0.03442 in the first epoch. Although the validation loss steadily decreased, significant improvement required multiple epochs, indicating that the model took longer to stabilize and begin effectively learning the underlying data patterns. In contrast, the LSTM model with sentiment features began with a slightly lower training loss of 0.0516 and a markedly lower validation loss of 0.00444 in the first epoch. This rapid initial decrease suggests that sentiment features provided immediate, meaningful context, enabling the model to grasp key patterns more effectively from the outset.

As training progressed, the convergence behavior of the two models diverged notably. The model without sentiment features reached its minimum validation loss of 0.00175 by epoch 11, after which the validation loss began to fluctuate. This fluctuation indicated challenges in generalizing to unseen data, likely due to the absence of contextual sentiment information,

leading to potential overfitting. Conversely, the sentiment-enhanced model continued to improve its validation loss until epoch 41, achieving a lower minimum validation loss of 0.00099. This prolonged and refined learning process demonstrates that sentiment features contribute to a deeper and more nuanced understanding of the data, allowing the model to continue enhancing its performance over more epochs without prematurely succumbing to overfitting. Additionally, post-epoch 41, the validation loss of the sentiment-enhanced model stabilized with minimal fluctuations, indicating better generalization capabilities and reduced overfitting compared to the non-sentiment model.

Quantitatively, the inclusion of sentiment features resulted in significant improvements across multiple evaluation metrics, including RMSE, MAE, and R^2 , for both training and testing sets. These metrics indicate that the sentiment-enhanced model outperforms the non-sentiment model across all evaluation metrics. Specifically:

- **RMSE (Root Mean Square Error):** Lower RMSE values indicate that the model's predictions are closer to the actual values. The sentiment-enhanced model reduced the Training RMSE by approximately **26%** and the Testing RMSE by **25%**, demonstrating improved prediction accuracy.
- **MAE (Mean Absolute Error):** Lower MAE values signify fewer average errors in predictions. The sentiment-enhanced model achieved a reduction of about **24%** in Training MAE and **20%** in Testing MAE, highlighting enhanced precision in its forecasts.
- **R^2 (R-squared):** Higher R^2 values reflect a greater proportion of variance in the dependent variable that is predictable from the independent variables. The sentiment-enhanced model increased Training R^2 by **9%** and Testing R^2 by **3%**, indicating a better fit and stronger predictive power.

The significant decreases in both RMSE and MAE, along with the substantial increases in R^2 values, highlight the enhanced accuracy and robustness of the model when sentiment data is incorporated. Lower RMSE and MAE values indicate that predictions are closer to actual values, while higher R^2 values reflect an improved proportion of variance explained by the model, indicating better predictive performance and reliability in the sentiment-enhanced model's predictions.

In terms of predictive performance, both models were capable of identifying general upward trends in closing prices. However, the model without sentiment features struggled with precision, particularly in capturing rapid fluctuations and significant price movements during volatile market conditions. This limitation was evident as the non-sentiment model tended to lag or miss substantial price changes, reducing its effectiveness in dynamic market environments. In contrast, the sentiment-enhanced model not only captured overall trends but also accurately detected sudden peaks and troughs. By incorporating sentiment data, the model could anticipate and react to market-moving news and shifts in investor sentiment, enabling it to adapt predictions in real-time and maintain closer alignment with actual prices even during periods of high volatility.

The ability to mitigate overfitting was another critical area where the sentiment-enhanced model outperformed its counterpart. The non-sentiment model exhibited signs of overfitting, as indicated by fluctuating validation loss after epoch 11, suggesting that it was learning noise and specific patterns in the training data that did not generalize well to new data. In contrast, the sentiment-enhanced model showed a more consistent validation loss with minimal fluctuations, indicating reduced overfitting and greater robustness. The inclusion of sentiment features acted as a form of regularization, introducing additional meaningful variability that helped the model generalize better to unseen data.

Regarding computational efficiency, the sentiment-enhanced model required more training epochs (**41 epochs**) to reach optimal performance compared to the non-sentiment model, which achieved optimal performance by epoch **11**. While this extended training duration implies a longer computational effort, the substantial gains in predictive accuracy and generalization justify the additional time invested. The improved model efficiency, characterized by lower losses and error metrics, ensures that the computational resources are utilized more effectively in the long run, providing a more reliable and accurate tool for stock price prediction.

In summary, the integration of sentiment features into the LSTM model markedly enhances its predictive capabilities. The sentiment-enhanced model achieves significantly lower RMSE and MAE values, indicating higher accuracy, and demonstrates improved stability and generalization, as evidenced by more stable validation loss and increased R^2 values. Additionally, the model exhibits superior ability to handle market volatility and capture nuanced market signals driven by investor sentiment and news impact. Although the sentiment-enhanced model requires longer training periods, the benefits in terms of predictive

performance and robustness make it a more effective and reliable tool for stock price prediction. This comparative analysis clearly demonstrates that sentiment features are a valuable addition to traditional numerical inputs, offering a more nuanced and accurate framework for understanding and anticipating market dynamics.

Discussion

The integration of sentiment analysis into stock price prediction models significantly enhanced the performance of LSTM models in forecasting. By incorporating sentiment features derived from financial news headlines, the LSTM model achieved substantially lower Root Mean Square Error (RMSE) values on both the training and testing datasets compared to a model without sentiment inputs. Specifically, the training RMSE dropped from 614.20 to 454.12, while the testing RMSE decreased from 337.48 to 253.78. These improvements highlight the value of sentiment data in capturing complex market behaviors that are not fully represented in numerical data alone. The results confirm the importance of including sentiment features to better reflect investor sentiment and market reactions.

A reliable data collection pipeline was key to this study's success, enabling the collection of financial news headlines from multiple sources to create a diverse and unbiased dataset. Tools like Selenium and BeautifulSoup ensured efficient scraping, while preprocessing addressed issues like missing data and redundancy. The creation of sentiment features, including lagged sentiment values and sentiment volatility indicators, allowed the model to capture both short-term reactions and long-term patterns in market sentiment. This comprehensive feature set contributed to the model's improved performance, as demonstrated by lower validation losses and RMSE values. Aggregating sentiment data from multiple financial news outlets reduced biases that are often present when relying on a single source. This diversity provided a richer context, helping the model generalize better to unseen data. Additionally, using sentiment data from reputable financial news sources ensured that the analysis was grounded in reliable and relevant information, further improving the model's predictive accuracy.

State of the Art

This study aligns with existing research showing that sentiment analysis improves financial forecasting. Previous studies, such as Ding et al. (2015), demonstrated that combining news text with stock price data enhances prediction accuracy for individual stocks.

Similarly, Hu et al. (2018) integrated sentiment analysis into LSTM networks, finding significant improvements in predicting stock trends. However, this study differs in several key aspects. While most prior studies focus on predicting individual stock prices or rely on sentiment data from a single source or social media platforms, this research targets the Dow Jones Industrial Average (DJIA), a major index that aggregates 30 companies across diverse industries. Index prediction introduces unique challenges compared to individual stock prediction, as it requires capturing the collective movement of multiple companies, each with varying levels of influence on the index.

Unlike individual stock predictions, which are often driven by company-specific events, predicting an index demands advanced techniques to account for the aggregate impact of news and sentiment. This study addressed these complexities by introducing sophisticated feature engineering techniques, including lagged sentiment values and sentiment volatility indicators, to capture both immediate reactions and long-term trends. These features allowed the model to account for temporal relationships in sentiment and price movements, offering a level of detail beyond basic sentiment scoring methods used in earlier research like Nelson et al. (2017).

A key innovation of this study is its aggregation of sentiment data from multiple financial news sources, reducing the biases associated with single-source analysis. By collecting data from several reputable outlets, the model was able to capture a broader spectrum of market sentiment, improving prediction accuracy. Additionally, integrating sentiment features with traditional market indicators in an LSTM framework tailored for index prediction is a relatively unexplored approach. This combination of qualitative and quantitative data proved effective in addressing the unique challenges of index-level forecasting.

In summary, this study advances the state of the art by tackling the complexities of index-level prediction, employing sentiment aggregation from diverse sources, and leveraging advanced feature engineering. These contributions highlight the novelty of the approach and its ability to enhance predictive accuracy for stock indices like the DJIA.

Critical Analysis and Limitations

Although the incorporation of the sentiment features enhanced the model performance, a number of limitations were still evident. Using only headline news for sentiment analysis may fail to capture the contexts of the articles. The headlines often involve sensational language in order to draw a lot of attention, which might not truly suggest the underlying

content. Using such a sentiment score technique reduces simpler emotions into positive or negative values and may not capture most of the nuanced investor reactions. These shortcomings could be surpassed by using some more sophisticated NLP techniques—for example, a deep learning-based sentiment analysis, or the text analysis of whole articles.

The model also suffered during periods of high market volatility; thus, other factors must be incorporated, including macroeconomic variables, geopolitical crises, and sudden disasters. These always cause sudden changes in the market, which cannot be portrayed in real time by festive sentiment or historical price movements.

The other challenge was weighting the sentiment data of individual companies in DJIA. Larger companies with larger market shares have more influence on the index, and yet, simple averaging may not reflect this appropriately. Future studies could look into dynamic weighting schemes based on market capitalization or trading volume to ensure that dominant companies are proportionally represented about their sentiments.

Implications of the Study

This work contributes considerably to theoretical research and practical application. Theoretically, it underlines the urgency of using qualitative data, like investor sentiment, to complement traditional numerical indicators. The complementary importance of finance, data science, and Psychology has provided a broader depth in understanding market behavior. It provides a challenge to the traditional reliance on models that use only quantitative data. This, in turn, has ramifications from a practical point of view for investors, traders, and financial analysts, especially those involved in whatever form of index-based investment or some form of derivative product. Models with sentiment analysis embedded can capture more insight into market behavior and professionals who can make better decisions with regard to index funds, ETFs, and index futures. This skill could reinforce various risk management strategies and lead to superior investment performance, particularly in markets where changes in sentiment are known to strongly influence prices.

Conclusion and Future Work

Conclusion

The integration of sentiment analysis with stock index prediction models has shown much potential for improving such models' accuracy, robustness, and overall performance. This research applied this concept by incorporating features related to sentiment extracted from financial news headlines, which significantly improved the prediction performance of the DJIA. The LSTM, with sentiment features, consistently outperformed the baseline model, with considerably lower RMSE and higher values of R^2 . These represent the power of qualitative data in capturing intricate response dynamics that may remain elusive to traditional quantitative stale indicators. This study confirms that investor sentiment, as derived from various financial news sources, plays a pivotal role in predicting the movement of stock indices, hence providing insights that enhance the interpretability and responsiveness of predictive models.

Advanced feature engineering techniques included in the model were volume-weighted sentiment, EWMA, and sentiment volatility, which helped capture short-run and long-run dynamics in market behavior. Volume-weighted sentiment ensured that the impact of the sentiment of each firm on the index was in proportion to its trading activity. This, therefore, correctly captures the dynamics of the market, while EMWA, with its smoothing nature, helped the model identify whether it is a noisy fluctuation or a change in sentiment trends. Similarly, volatility in sentiments picked up the variation in investor sentiments and framed the periods of market uncertainty. These features enabled a more nuanced understanding of market sentiment, especially during periods of heightened volatility, when accurate predictions are most critical.

Sentiment aggregation across news sources proved instrumental in minimizing the biases inherent in single-source analyses. The model used a broad spectrum of sentiment data, meaning its view of market sentiments was richer and more comprehensive, hence generalizing and being more robust. This is useful in bringing forth diverse investor sentiment that helps to minimize the over-reliance on a particular fraction of a story that may bias predictions. From these results, integrating sentiment from multiple reputable sources is crucial in building more resilient financial forecasting models.

However, several limitations need to be considered. Although effective, all this relies on news headlines for sentiment scoring at the risk of simplifying investor emotions and omitting critical nuances in the full text of news articles. Headlines are often drafted to seize attention and may sometimes fail to convey the detailed context necessary for comprehensive sentiment analysis. The very sentiment scoring mechanism that mainly reduces the emotion to three levels-positive, neutral, and negative-may turn out quite crippling in being able to capture the full spectrum of investor emotions and their subtleties.

Furthermore, there are still challenges in giving the proper weight to individual company sentiment in the DJIA. The methodology may not accommodate the disproportionate influence given to larger companies. More dynamic weighting schemes could be developed with considerations like market capitalization, trading volume, or sector-specific influences to yield more generally accurate representations of market sentiment. These various refinements will ensure that the sentiment of the companies with substantial market influence is captured suitably, hence improving the predictive power of the model.

This is also where future studies need to be cognizant of the potential for alternative sources of sentiment: social media platforms, financial forums, and real-time investor feedback. These latter three sources go hand in glove to provide valuable, timely inputs found in public sentiments that may have very momentous consequences for changes in market behavior. Second, the deep learning architecture could utilize more advanced mechanisms, such as attention mechanisms or transformer models, so that it may allow the model to zero in on the most relevant segments of the input data and make predictions at a higher degree of accuracy, especially during periods of turmoil in the market. Further, applying macroeconomic indicators with sentiment features can also scale up the predictive capability by considering a holistic understanding of the driving factors behind market movements. Finally, this research adds one more critical building block to the growing volume of evidence that supports the integration of sentiment analysis into financial prediction models. This paper clearly shows the practical value derived from the crossroads between quantitative and qualitative data in developing better methods for financial modeling. Integrating diverse sentiment features has shown promise in improving accuracy, responsiveness, and robustness in stock market predictions. Moving forward, integrative data, dynamic weighting methodologies, and increasingly advanced model architectures must be fully utilized to further improve predictive models in financial markets.

Future Work

From this, several recommendations arising from the results relate to future research and practice:

Incorporating Alternative Sentiment Sources: Extending the sentiment analysis to platforms like Twitter or even to Reddit, financial forums, and company announcements could keep real-time investor sentiments and, therefore, make the model responsive to quick changes in the market. Social media has been noted to affect the behavior of the markets and can even offer up-to-date reflections of public opinions.(Kaminski et al., 2018).

Experimenting with Advanced Architectures: Integrating attention mechanisms or transformer models could allow the model to focus on the most relevant information in the data, potentially improving predictions during volatile periods. These models have shown success in natural language processing tasks and could enhance the model's ability to interpret complex sentiment patterns.

Incorporating Time-Sensitive Sentiment Analysis: Accounting for the timing of news events relative to trading hours might make the model more capable of forecasting immediate price reactions. Markets can show distinct reactions to news released during trading hours versus after hours and capturing this temporal aspect could improve the prediction accuracy.

Combination with Macroeconomic Indicators: It will enable the acquisition of wider knowledge of market conditions by integrating macroeconomic indicators, sector-specific news, or technical analysis indicators into its interest rate and unemployment rate data. Such a multimodal approach acknowledges that the various stock indices are a result of a number of different factors that are going on, and incorporating diverse sources of data might provide a more comprehensive overview.

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